Alo Roosing

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Experience

Esri R&D Center Zürich
- *June 2020 – April 2023* **Software Engineer**
  Projects included atmospheric scattering, volumetric clouds, OIT and particles.
  Technologies: WebGL, GLSL, TypeScript

- *November 2019 – April 2020* **Software Engineer Intern**
  6-month internship developing volumetric cloud rendering.

Swiss National Supercomputing Centre
- *February 2019 – October 2019* **Scientific Software Developer**
  Technologies: C/C++, CUDA, MPI

Education

University of Cambridge
- *2015 – 2019* **PhD Scientific Computing**
- *2013 – 2014* **MPhil Scientific Computing**

King's College London
- *2010 – 2013* **BSc Computer Science**

Skills

I have professional experience implementing methods for **real-time rendering** using **WebGL** and **GLSL**. This has included authoring **shaders** for physically based rendering (**PBR**) and **volumetric** effects.

I have worked in large teams using **Git** for version control and written automated **tests** for multi-platform products. During my studies I worked with **GPGPU**s to **research** and implement fluid **simulation** pipelines. I also have historical experience using **C/C++** and **Linux**.

For my hobby projects I have explored **procedural generation** and **screen space effects**. I have experience implementing graphics standards and publications (e.g. GLTF 2.0 specification, GGX model).

Publications

Fast Distance Fields for Fluid Dynamics Mesh Generation on Graphics Hardware
A. Roosing, O. Strickson, N. Nikiforakis, Commun. Comput. Phys., 26 (2019), pp. 654-680.

Graphics Programming Portfolio

Alo Roosing

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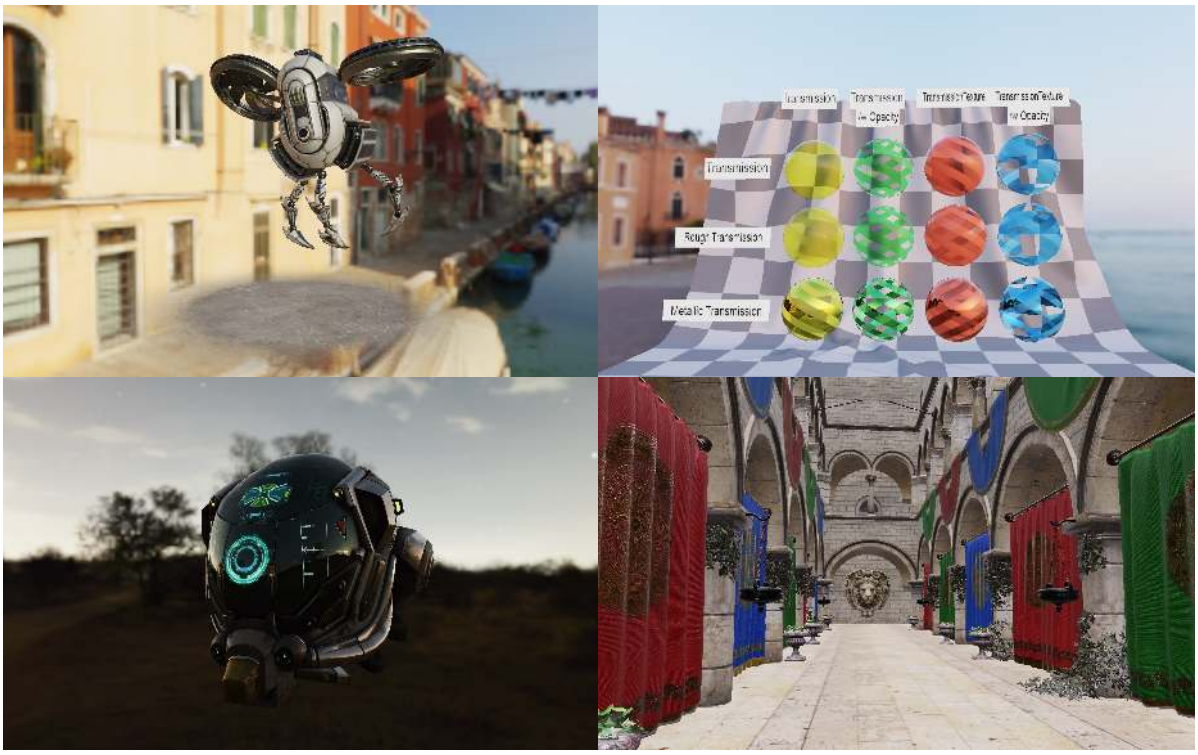
WebGL Renderer

Live demo: https://al-ro.github.io/projects/renderer/

Technologies: WebGL2, GLSL ES 3.0, JavaScript, GLTF 2.0

As a hobby project, I have been developing a WebGL2 renderer. It is a flexible real-time physically based forward renderer based on the GLTF 2.0 specification. This project is WIP.



The renderer supports environment based lighting, animation, transmission, emission and other features

GLTF 2.0

GLTF 2.0 is a popular asset transfer format which allows PBR models to be shared in an easy and compact way. The project currently supports: attribute and index extraction (sparse and dense), traversing nodes and primitives, PBR material extraction (with properties and textures), animations, morph targets and skinning. The engine also implements the transmission, IOR, sheen and texture transform extensions.

Importing data

We traverse and process the GLTF scene graph of nodes, generate parameters for the internal engine representation and allocate GPU buffers for mesh attributes. An internal engine node graph is created which represents the hierarchy of the scene. Some nodes are drawable mesh objects which correspond to geometry, morph and skinning data coupled with a shader program.

To avoid data and work duplication, the engine keeps track of buffer descriptions and shader programs in repository maps. As different primitives are not aware of shared data, whenever they seeks to create an already allocated buffer or a previously compiled shader program, we can reuse the existing ones. This helps cut down on GPU memory use and shader compilation times.

Materials

In the engine, materials correspond to GLSL programs. Each drawable primitive is created with a PBR material which is generated to exactly match only the features and aspects that the primitive has. The engine material system is extendible and modular and there exist other materials such as textured, attribute visualising or Lambertian ones. There are also materials for the environment, the generation of IBL data and a depth pre-pass. Shader authoring and integration is simple and new materials can be added to the engine.

Physically Based Rendering

The engine uses physically based rendering (PBR) and image based lighting (IBL). We use a Lambertian diffuse lobe and the Cook-Torrance BRDF for the specular lobe. We use the Trowbridge-Reitz (GGX) microfacet distribution function, the Smith visibility/shadowing function and the Schlick Fresnel approximation.

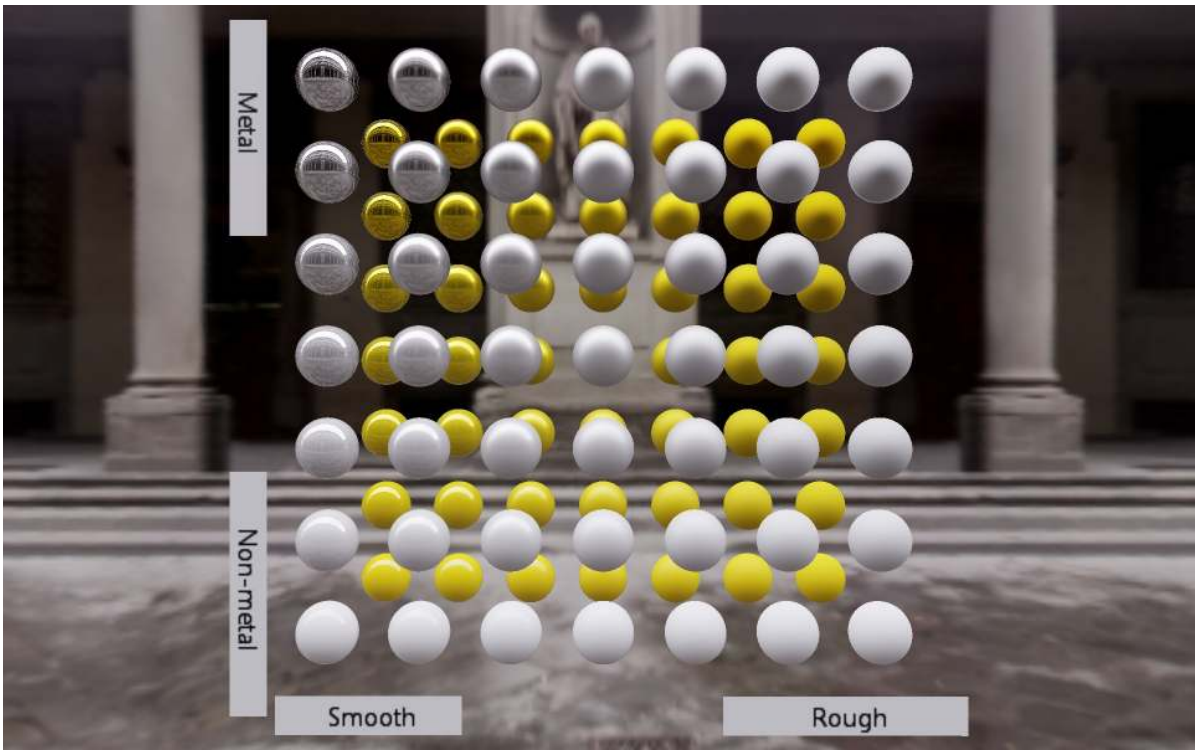
HDR environment maps act as the source of illumination. Whenever a new HDR environment texture is selected, a pre-processing program convolves the environment data and

generates spherical harmonics matrices for diffuse ambient light. Another shader generates a cubemap used for ambient specular light. The lower mip-levels store pre-integrated specular reflections following Frostbite and Filament. We also generate a BRDF LUT as described in those approaches. The red and green channels hold the scale and bias for the GGX BRDF and the blue channels holds the DG term for the Charlie BRDF used for rendering sheen.

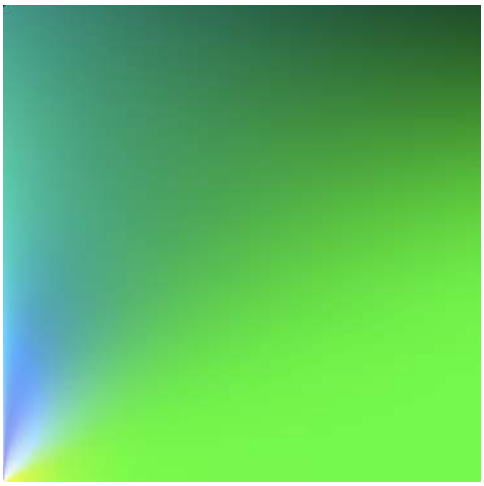
The renderer makes use of data and colour textures to render spatially varying materials illuminated by the environment.

Render order and performance

Although the primitives exist in a common node graph, we distinguish between opaque, transmissive and transparent meshes and use render-passes to achieve correct blending. The PBR materials can be quite heavy for a forward renderer in a browser. We use AABB culling to exclude primitives which fall outside of the camera view frustum. This takes into consideration node animations and updates the bounding volumes whenever anything in the scene changes. We also use a depth pre-pass of all opaque geometry to avoid doing useless work in the fragment shader which would be overdrawn by subsequent primitives.

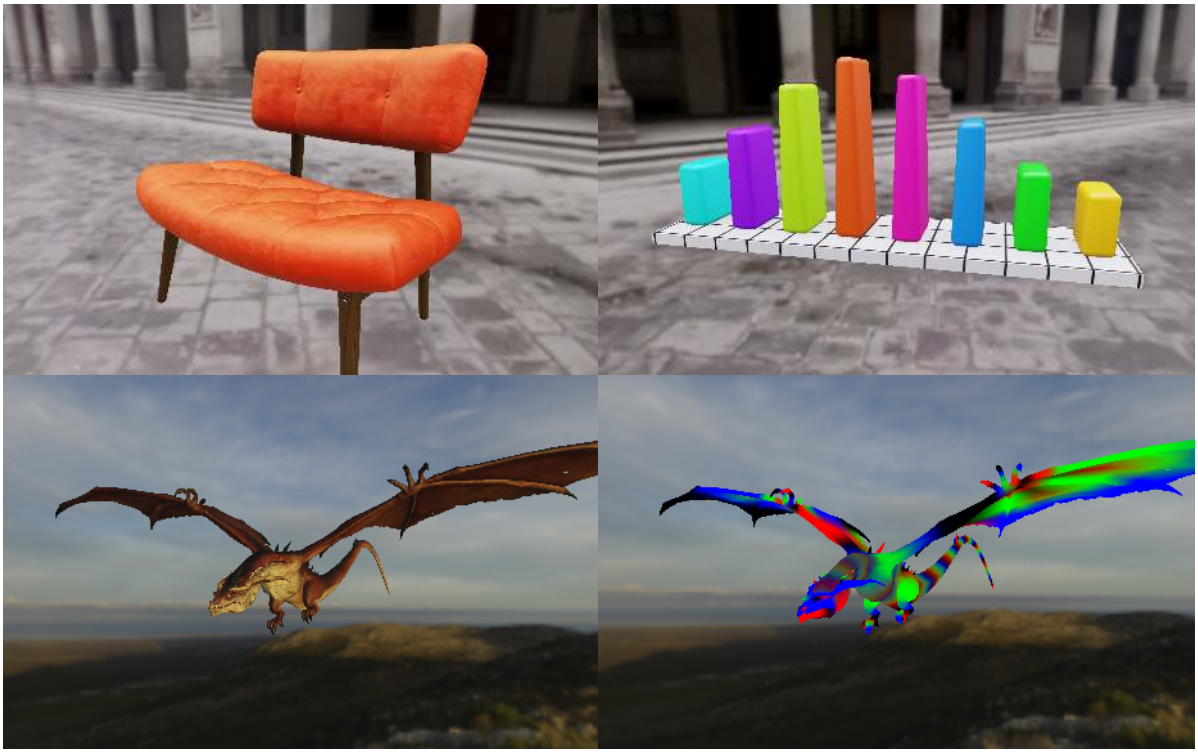


The mip-levels of the environment cubemap store convolved specular reflections for varying roughness. Diffuse illumination is stored as spherical harmonics matrices.



The renderer uses the popular split-sum approach for image based lighting. The BRDF LUT hold data which depends only on the view direction and roughness. Note that is the multiple scattering variant based on Filament.

Morph targets and skinning use data textures to avoid attribute limits. Morphed geometry has a texture array where each level is a square texture storing the interleaved attribute offset data for one target. Skins store and update a texture which holds the transform matrices of their joints.



The renderer supports sheen, unlimited morph targets and skinning. Vertex skin joint influence visualised in the last image.

Summary

As a learning experience, the hobby renderer has been an invaluable tool which has highlighted the need for flexibility and avoiding assumptions. The implementation has relied on publicly available documentation, tutorials and example code. Writing a standalone renderer from the ground up has included research and development of multiple interacting features and raised interesting questions about design and performance aspects.

ArcGIS JS API

Live demo: https://developers.arcgis.com/javascript/latest/sample-code/scene-weather/

Technologies: WebGL2, GLSL ES 3.0, TypeScript

As part of the WebGL engine team at Esri R&D Center, Zurich, I helped develop and maintain the ArcGIS JavaScript API. Our team was responsible for the 3D rendering engine used for displaying terrain, buildings, visualisation layers and many other effects to convey information about both local and worldwide projects and phenomena. My focus was on improving the visual fidelity of the rendering software. I was the main developer on a new realistic atmosphere, cloud rendering and weather effects. I also added order independent transparency and helped test, debug and improve the codebase.

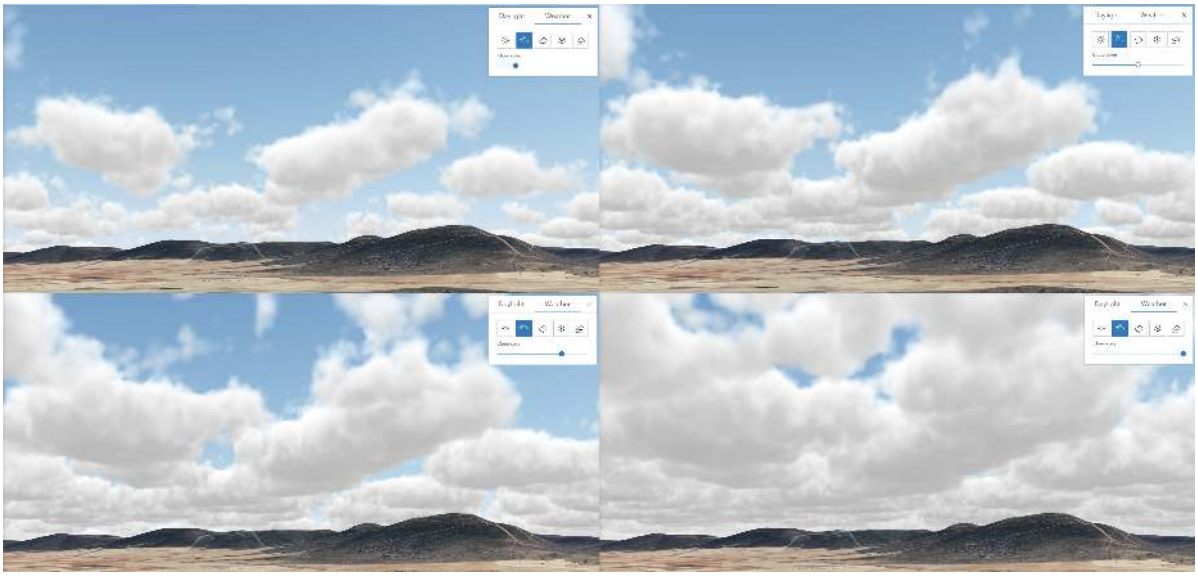
Atmospheric Scattering

The new adaptive resolution atmosphere is visible from ground level and global views.

The atmosphere implementation is based on Schüler's approximation of the Chapman function. The linear depth of the scene is used to ray march the atmosphere for each pixel in a screen space quad. At each sample point, the extinction of light from the sun is found using the absorption and in-scattering along the light ray. The colours are achieved by simulating Rayleigh scattering with Mie scattering around the light direction to add a haze around the sun. The sun itself is rendered as a simple directional gradient on the screen space quad but the colour also makes use of light attenuation to achieve reddish tones near the horizon.

The design aimed for a balance between aesthetics and data preservation. As JS API is used for data visualisation, only in-scattering along the view ray is applied, ignoring absorption which would hide the basemap and geometry. Of particular interest was that user interaction not be interrupted for rendering work which led to several performance optimisations. Only a few ray marching steps are used and when the user interacts with the scene, the atmosphere is rendered at a smaller resolution and interpolated across the screen. Once user interaction has ended, a full resolution image is rendered and displayed.

Volumetric Clouds



Cloud coverage is controllable. Changes trigger render of the cubemap.

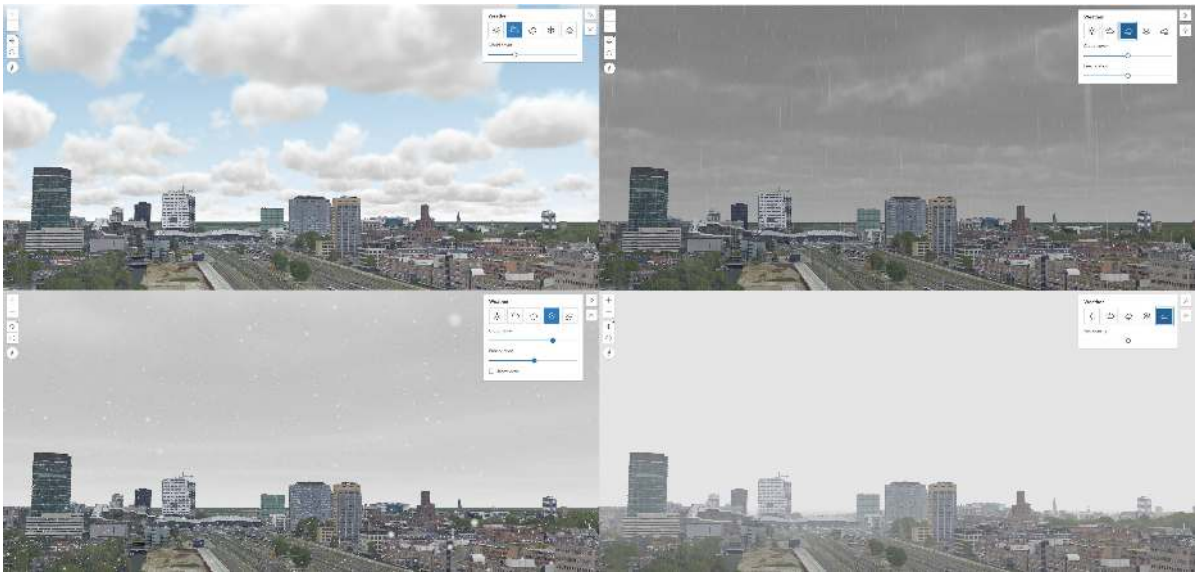
We implemented cloud rendering based on the work by Schneider and Guerrilla games. We ray march a shell of clouds around the camera, keeping track of light attenuation. The base density is set by a cloud map texture which is generated procedurally based on the location of the camera and the date of the scene. A Gradient-Worley noise volume is generated and stored

as a 2D atlas with transmissive halo cells around each tile. This volume data is used to carve details from the base density and generate the final cloud shape. The clouds are lit by sampling a light direction from directly above for each sample point combined with a height-based ambient term.

Because real-time rendering of volumetric clouds is prohibitively expensive for the JS API (which works in browsers), the effect is achieved with a cubemap which stores the colour values from the ray marching and the total transmittance along the view ray. Five faces are generated across five frames whenever there is a change to the variables of the clouds, the position of the camera or the date of the scene. Once this has finished, we use the global sun direction and colour to add directional shading to the cubemap.

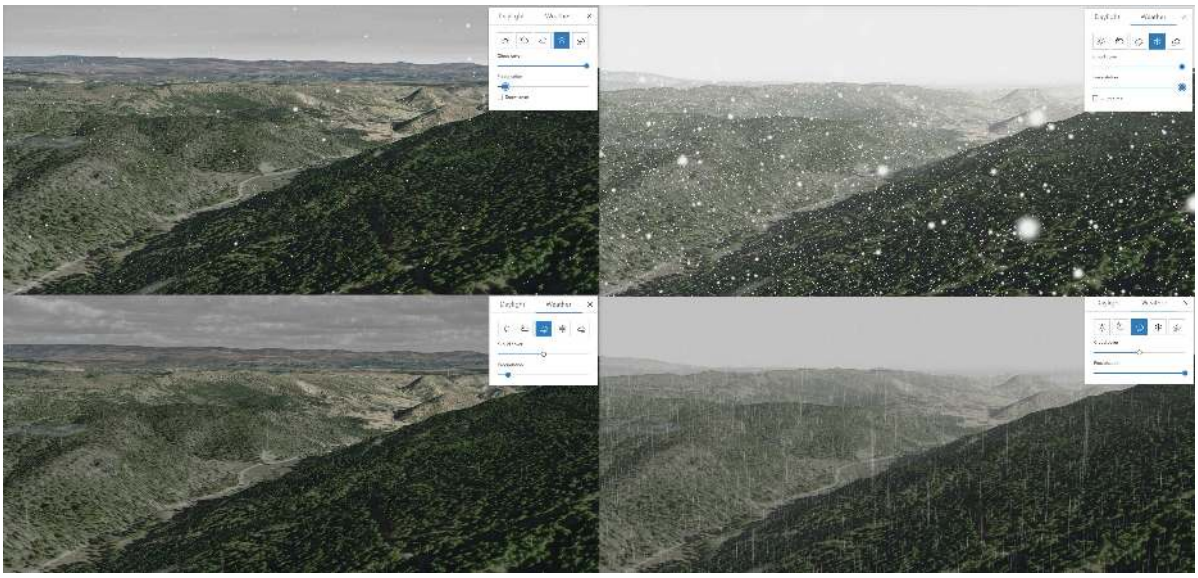
From a design point the project included a lot of discussions with several stakeholders about the desired look of the clouds and the level of control our developers and the user would have. We developed a fully controllable system but only exposed a set number of parameters in the internal code. From a technical standpoint, it was an interesting challenge to get the solution to work and perform well for low-end devices such as older integrated graphics cards and mobile phones. This often required dynamic testing of device capabilities and changing the complexity, number and order of operations in GLSL shaders.

Weather Effects



Weather types allow the user to set the mood of the scene.

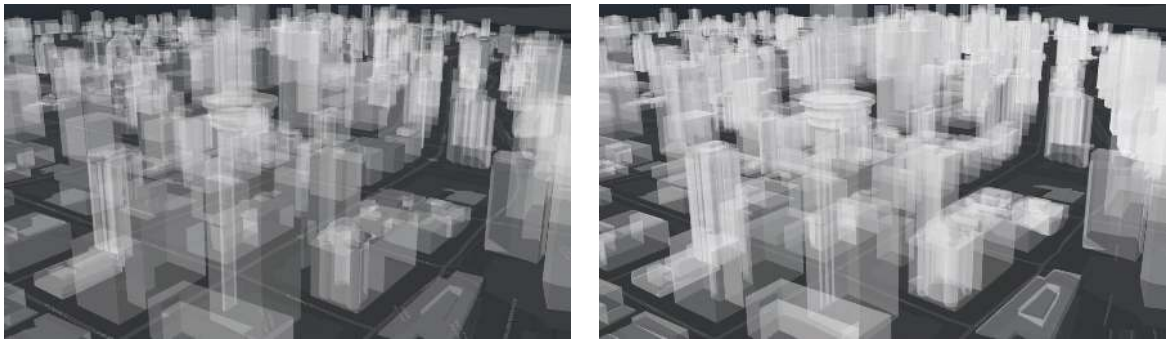
The cloud system was extended to implement a weather system. There are several weather types which make use of the clouds with different variables. The system added particle rendering for rain and snow effects and a distance fog. The particles are instanced triangles which are rendered around the camera in world-space and fall towards the centre of the globe, disappearing behind world geometry. As the camera moves, the particles remain static relative to their world-space positions allowing the viewer to fly through the precipitation. The camera is surrounded by an implicit volume and as particles exit one side, they are placed at the other boundary, creating the illusion of infinite raindrops.



Setting the strength of the precipitation controls the number of rendered particles. Depth fog is added to simulate distant particles.

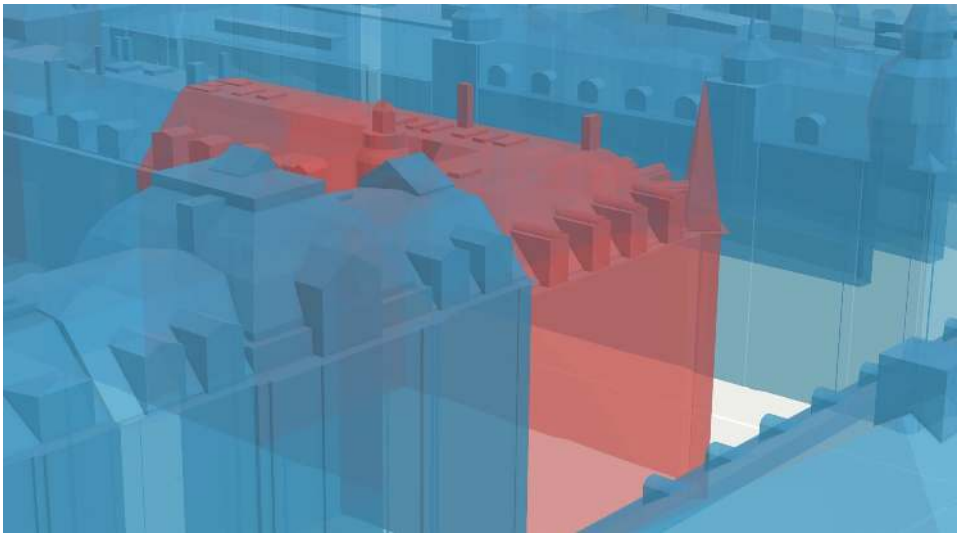
Order Independent Transparency

Order independent transparency is important to JS API as it seeks to represent comparative spatial information. The implementation follows the approach by McGuire without the weightings. The use case for cityscape scenes is difficult for weighting as there are many distant clusters of closely positioned geometry. We render accumulated colour and coverage into two floating-point textures and mix the result with a third image which contains the front-most faces of all transparent geometry. This allows us to differentiate between the relative depths of the transparent meshes.



a) Depth test culling b) Order Independent Transparency

OIT allows to render transparent geometry without sorting.



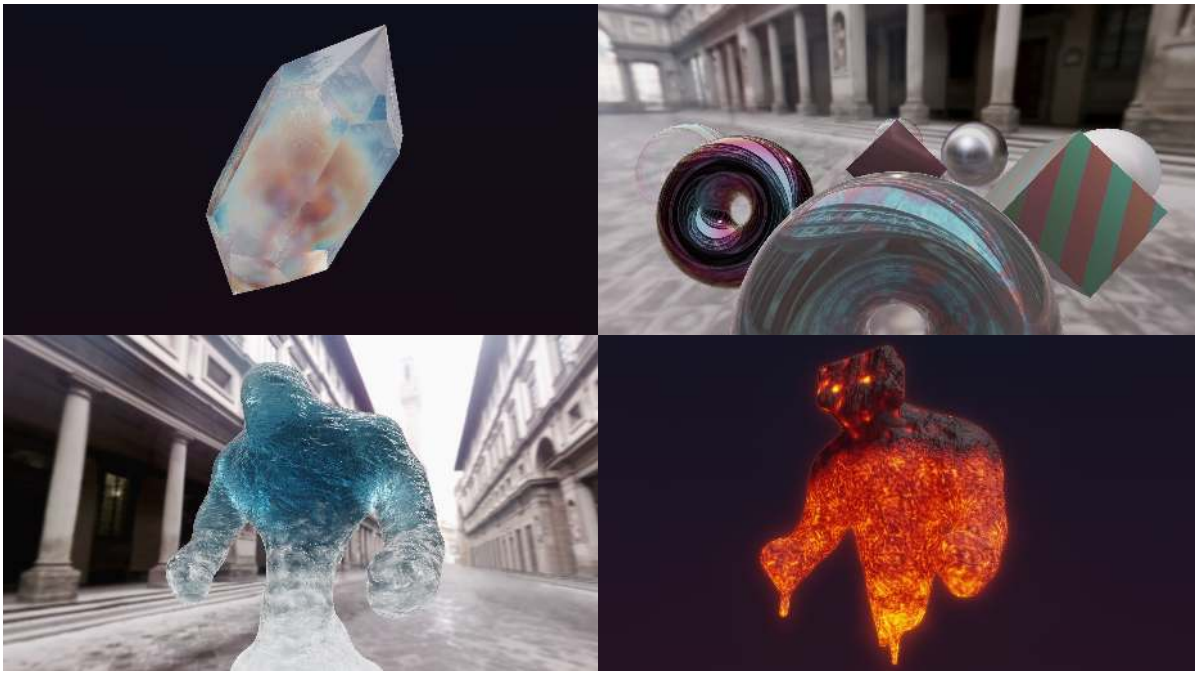
The implementation allows the viewer to differentiate between relative depths.

Shadertoy

Live demos: https://www.shadertoy.com/user/alro

Technologies: GLSL ES 3.0

Shadertoy offers an online fragment shader platform and has its origins in the size- and feature-limited graphics of the demoscene. Shadertoy is a great place for prototyping and making artistic shaders. The community interaction and collaboration help foster an environment where anyone can learn. The user has access to a single screen space quad, several preset textures, multiple render targets and some uniforms.

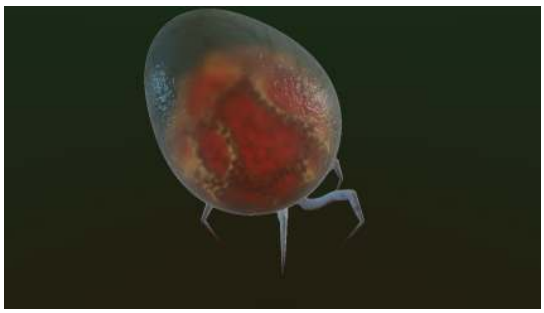


Selected screen space shaders. Featuring volumetric effects, iridescence, refraction and black-body radiation.

Case Study

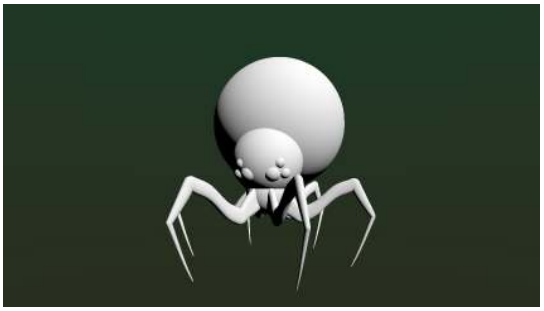
As an illustration of the work that has gone into these shaders, let's analyse the "Swamp Stalker". The shader renders a 3D creature with translucent skin and a partly transparent body. It features subsurface scattering, normal mapping and rough transmission to achieve a generally unsettling organic look.

Creature with subsurface scattering

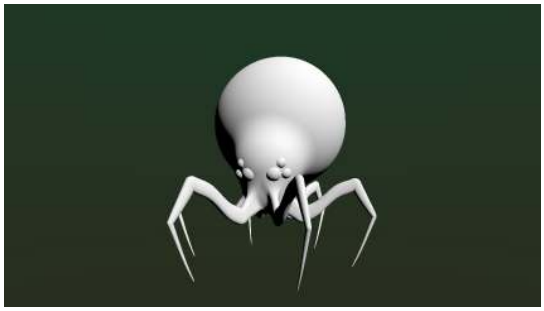


Depth varying rough transmission

The vertex shader is just two triangles which cover the screen and all work is executed in the fragment shader. We begin by generating camera rays for each fragment and determine the ray-geometry intersection. We do this by ray marching the signed distance functions of geometric primitives like spheres and capsules. By using constructive solid geometry approaches such as smooth minimum and intersection functions, we can achieve complex shapes. The legs use radial repetition to create several instances of a geometry while only paying the cost of evaluating a single copy. We also warp space for the legs to bend straight primitives into more organic shapes. The sections of the body are assigned bounding volumes to reduce work at sample points and increase performance.



Ray marching geometric primitives

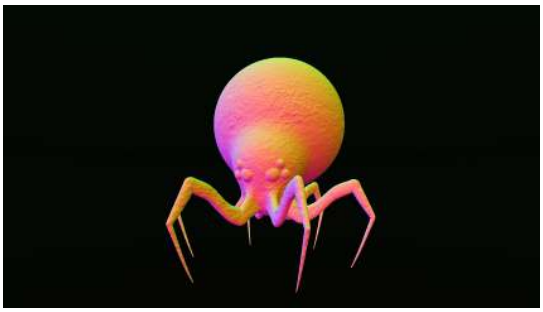


Applying smoothing functions

Knowing the intersection point allows us to do shading. We can determine the normal of the surface by sampling the distance function value in the neighbourhood of the intersection and finding the local gradient. Knowing the surface normal, we can generate a local orthonormal

coordinate system and apply normal mapping. We do this with a noise texture using tri-planar mapping and finding the detail gradient around the shading point.

Geometry normal



Normal mapping

We layer sine waves to generate the albedo of the surface and use two lights to apply PBR shading. The specular contribution has a direct part from the lights and an ambient part based on a simple reflection and gradient.

Albedo applied

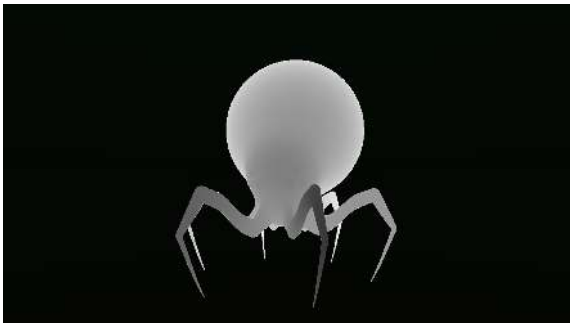
PBR shading

Direct specular

Ambient specular

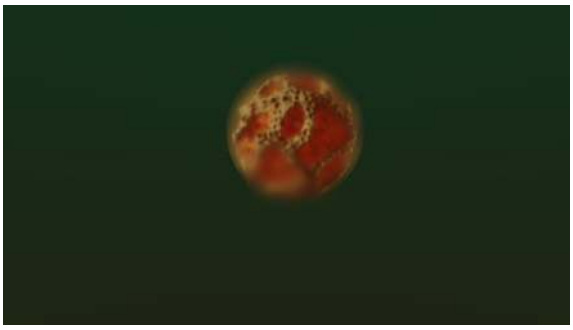
For the internal parts, we render a scene of a layered gyroid in a separate pass. We generate mipmaps for this new texture and use bi-cubic interpolation to mix values and achieve controllable smooth blurring in a single pass. We use the transparency of the body and the differences in depths to mix the two scenes.

Internal geometry

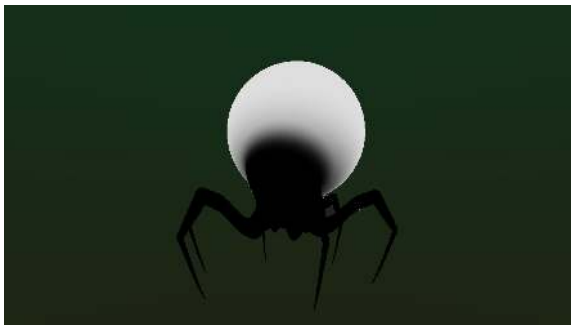


Depth of main scene

Depth difference



Blurring based on difference

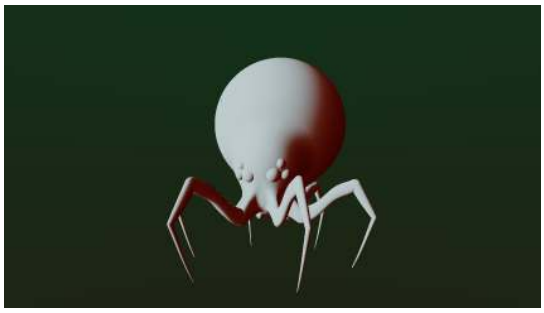


Transmission gradient

Combined with no SSS

To achieve a good flesh effect, we need to render subsurface scattering (SSS) as skin is translucent. When light enters skin, it scatters and exits some distance away, having been attenuated in different wavelengths. We use a pre-integrated screen space SSS approach. We generate a LUT which encodes the scattering values with surface normal and light direction alignment on one axis and the thickness of the material on the other. The SSS value is applied for both light sources and multiplied with the albedo and radiance to be the diffuse contribution of the illumination.

Pre-integrated SSS texture



SSS LUT for one light

SSS evaluation from both lights

SSS multiplied with albedo

Final result with transmission, lighting and SSS combined

Summary

Compared with the classical render pipeline, Shadertoy offers a limited set of features. However, much of the work is transferable to other paradigms and the focused nature of these small shaders helps investigate and optimise specific approaches. 3D scenes often use ray marching or ray tracing and care is required to achieve good performance. I have implemented several different rendering techniques and continue to share shaders with the wider community and learn from others and from my own investigations.

Selected screen space shaders. Featuring volumetric clouds, procedural materials and 2D effects.